Audit.sql

```sql
-- Tutorial 12.3, test H8: days when everyone clocked the same hours
-- Population: manufacturing clock entries; expected: no such days
WITH PlantDays AS (
    SELECT tc.WorkDate, COUNT(DISTINCT tc.EmployeeID) AS Employees,
        COUNT(DISTINCT tc.RegularHours + tc.OvertimeHours)
            AS HourPatterns,
        SUM(tc.OvertimeHours) AS OvertimeHours
    FROM TimeClockEntry AS tc
        INNER JOIN Employee AS e ON e.EmployeeID = tc.EmployeeID
        INNER JOIN CostCenter AS cc ON cc.CostCenterID = e.CostCenterID
    WHERE cc.CostCenterName = 'Manufacturing'
    GROUP BY tc.WorkDate
)
SELECT strftime('%Y', WorkDate) AS WorkYear,
    SUM(CASE WHEN HourPatterns = 1 AND Employees > 1
        THEN 1 ELSE 0 END) AS SurgeDays,
    MIN(CASE WHEN HourPatterns = 1 AND Employees > 1
        THEN Employees END) AS EmployeesPerDay,
    ROUND(SUM(CASE WHEN HourPatterns = 1 AND Employees > 1
        THEN OvertimeHours ELSE 0 END), 0) AS SurgeOvertime,
    ROUND(SUM(CASE WHEN HourPatterns = 1 AND Employees > 1
        THEN OvertimeHours ELSE 0 END) / SUM(OvertimeHours), 3)
        AS SurgeShare
FROM PlantDays
GROUP BY WorkYear
ORDER BY WorkYear;
```

	WorkYear	SurgeDays	EmployeesPerDay	SurgeOvertime	SurgeShare
1	2024	12	64	3456.0	0.355
2	2025	32	64	9216.0	0.589
3	2026	49	64	14112.0	0.705

Execution finished without errors.
Result: 3 rows returned
At line 451:
WITH PlantDays AS (

Outlined: the days on which every one of the 64 hourly manufacturing employees clocked the same hours, and their share of the year's manufacturing overtime.